STUDENTHUB

Student Record and Academic Management Platform

UNIT-2 AGILE MINI-PROJECT REPORT

Submitted by: _____

Class / Department: _____

College: _____

Academic Year: _____

1. Introduction

StudentHub is a web-based prototype designed to help an educational institution organize student details, course information, attendance and academic marks in one place. The project follows Agile planning concepts and delivers a small Minimum Viable Product (MVP).

2. Product Vision

To develop a simple, user-friendly and organized Student Record and Academic Management Platform that helps authorized educational users maintain student profiles, courses, attendance and academic performance efficiently.

3. Objectives

• Maintain student registration details in a structured format.

• Create and view course records.

• Record student attendance by course and date.

• Enter marks and calculate a basic grade.

• Display summary counts and allow JSON data export.

4. Target Users and Stakeholders

• Faculty members: manage student, course, attendance and marks information.

• Academic administrators: review and organize academic records.

• Students: are the subjects of the records and may view results in a future version.

• Development team: designs, builds, tests and documents the MVP.

5. MVP Scope and Constraints

The MVP includes student management, course management, attendance entry, marks entry, dashboard summaries and JSON export. This implementation is a browser-based educational prototype. It uses Local Storage and does not include a live server, central database, multi-user login or production security. Only sample data should be used.

6. Scrum Roles

• Product Owner: defines requirements, prioritizes backlog and accepts completed work.

• Scrum Master: facilitates Scrum activities, removes blockers and supports Agile practices.

• Development Team: designs the interface, writes code, tests features and prepares documentation.

7. Project Epics

Epic 1 – Student Management: Add student, view student records, search by name or roll number, delete student.

Epic 2 – Course Management: Add course, view course list and delete course.

Epic 3 – Attendance Management: Select student and course, choose date, mark Present or Absent, view records.

Epic 4 – Marks and Academic Performance: Enter marks out of 100, view marks and calculate a basic grade.

Epic 5 – Dashboard and Data Export: View summary counts and export stored data in JSON format.

8. User Stories and Acceptance Criteria

Student registration. As an administrator, I want to add a student so that student details are stored. **Acceptance:** The form accepts name and roll number; a saved student appears in the table; the dashboard count updates.

Course creation. As a faculty member, I want to add a course so that it can be used for academic records. **Acceptance:** Course name and code are required; saved course appears in the course list and selection lists.

Attendance recording. As a faculty member, I want to record attendance so that attendance information is maintained. **Acceptance:** A student, course, date and status can be selected; the entry is saved and displayed.

Marks entry. As a faculty member, I want to enter marks so that academic performance can be reviewed. **Acceptance:** Marks must be between 0 and 100; the result appears with a calculated grade.

Data export. As an administrator, I want to export records so that I can keep a backup. **Acceptance:** Export downloads a JSON file containing the available project data.

9. Prioritized Product Backlog

Priority	Backlog item	Story points
High	Add and view students	3
High	Add and view courses	2
High	Record attendance	5
High	Enter marks and grade	5
Medium	Search student records	2
Medium	Dashboard summary	3
Medium	Export JSON backup	3
Low	Advanced analytics and reports	8

10. Story Point Estimation

The project uses a simple relative scale: 1 = very easy, 2 = easy, 3 = moderate, 5 = difficult and 8 = very difficult. Story points represent relative effort and complexity, not hours.

11. Sprint Plan

Sprint 1 – Project setup and student management. Create repository and interface; implement add, view, search and delete student records; test student module.

Sprint 2 – Course and attendance. Implement course records; connect student and course selection; record and display attendance; test attendance flow.

Sprint 3 – Marks, dashboard and release. Implement marks and grade calculation; add dashboard counts and JSON export; test all modules; prepare README and final demonstration.

12. Scrum Board

Suggested workflow columns: Backlog → To Do → In Progress → Testing → Completed. Move each user story across the board as work progresses.

13. Scrum Ceremonies

• Sprint Planning: select backlog items, estimate effort and define the sprint goal.

• Daily Scrum: each member shares what was completed, what comes next and any blockers.

• Sprint Review: demonstrate completed features and collect feedback.

• Sprint Retrospective: discuss what went well, challenges and improvements for the next sprint.

14. Agile Framework Comparison

Framework	Main idea	Suitability
Scrum	Time-boxed sprints and defined roles	Good for a student team building modules in stages
Kanban	Visual workflow and continuous delivery	Useful for tracking small tasks
XP	Frequent testing, feedback and code quality	Useful for improving software quality
Lean	Reduce waste and focus on value	Useful for keeping the MVP simple
LeSS / SAFe / Scrum@Scale	Scaling Agile across multiple teams	More suitable for larger projects than this small MVP

Recommended approach: Scrum is suitable for this project because the work can be divided into clear sprints and reviewed after each increment. Kanban can also be used as a visual task board.

15. Testing

• Add a student with required details and confirm it appears in the table.

• Try searching using a name or roll number.

• Add a course and confirm it appears in the course dropdowns.

• Record Present and Absent attendance entries and confirm both are listed.

• Enter marks such as 85 and confirm the grade is calculated.

• Export JSON and verify that the file downloads.

16. Future Enhancements

• Secure login with role-based access.

• Backend API and central database.

• Student self-service portal and result viewing.

• Attendance percentage calculation and downloadable reports.

• Email or SMS notifications and audit history.

17. Conclusion

StudentHub demonstrates how Agile practices can be used to plan and develop a small academic management application. The prototype covers core student, course, attendance and marks workflows. Further development can add a secure backend, centralized database and advanced reporting.

18. GitHub Submission

Create a public GitHub repository, upload index.html, style.css, script.js, README.md and this report PDF. Enable GitHub Pages from Settings → Pages → Deploy from a branch → main → /(root). Copy the published website URL and repository URL into the Google Form as requested by the organizer.